

Teorema de Riesz-Fischer

Teorema 1 (de Riesz-Fischer). *Sea H un espacio de Hilbert y $\{u_n\}_{n \in \mathbb{N}}$ un sistema ortonormal. Entonces, la transformada de Fourier asociada a $\{u_n\}_{n \in \mathbb{N}}$ es sobreyectiva.*

Demostración: Sea $\bar{a} = (a_n)_{n \in \mathbb{N}} \in \ell^2$, definimos $\forall m \in \mathbb{N} : x_m = \sum_{n=1}^m a_n u_n$. Entonces, $(x_m)_{m \in \mathbb{N}}$ es una sucesión de Cauchy, pues para $k \in \mathbb{N}$:

$$\forall m \in \mathbb{N} : \|x_{m+k} - x_m\|^2 = \left\| \sum_{n=m+1}^{m+k} a_n u_n \right\|^2 \stackrel{2\text{Pitágoras}}{=} \sum_{m+1 \leq n \leq m+k} \|a_n u_n\|^2 = \sum_{n=m+1}^{m+k} |a_n|^2 \xrightarrow{m \rightarrow \infty} 0,$$

porque $\bar{a} \in \ell^2$. Como H es completo, existe $x \in H$ tal que $\lim_m x_m = x$.

$$\forall n \in \mathbb{N} : \hat{x}_n = \langle x, u_n \rangle = \left\langle \lim_{m \rightarrow \infty} x_m, u_n \right\rangle \stackrel{\text{prop-prod-interno-continua/??}}{=} \lim_{m \rightarrow \infty} \langle x_m, u_n \rangle = a_n,$$

ya que $\langle x_m, u_n \rangle = a_n$ si $m \geq n$ y 0 en caso contrario. Por tanto, $\mathcal{F}(x) = \bar{a}$. ■

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